



STACKING
WIZARD



ASTRO
WIZARD

USER MANUAL

Professional Astrophotography –
easy and uncomplicated.
Stack. Optimize. Enhance.



AUTOMATE

Simplify and automate
processes



STACK

Intelligent stacking
for maximum detail



OPTIMIZE

Improve and perfect
your captures



ENHANCE

Make results
brilliant

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Introduction and shared preparation

PART I - STACKINGWIZARD: Pile, Looking Glass, Stack, Handoff

PART II - ASTROWIZARD: detailed steps 1 through 9

Troubleshooting and final checklist

Program order is strictly separated. Every AstroWizard step includes purpose, controls, verification and new features.

Introduction and current release status

This manual supersedes the 31 August 2026 edition. It follows the complete workflow in the correct order: StackingWizard turns sub-exposures and calibration frames into a clean registered linear stack; AstroWizard then handles background, colour, sharpening, denoising, stretching and final presentation.

Verified release level: AstroWizard “True Colours” and StackingWizard Beta 4.4 “The Workbench”, both dated 18 September 2026. Development is rapid; for later builds, read the official changelog before relying on exact labels.

Major changes since the previous guide

Area	New level	Practical meaning
StackingWizard	Beta 2.4 → 4.4	Live Canvas for walks and mosaics, joint level solving, cloud verdict, Workbench controls, improved sorting and receipts.
AstroWizard colour	Star-True Colours	Gaia-based photometric colour calibration for suitable linear broadband data.
AstroWizard stretch	Manual GHS	Generalized Hyperbolic Stretch with shared histogram/curve, symmetry point and protection controls.
Dual-band	Two OSC stacks → SHO	Ha/OIII plus SII/OIII are registered, separated and prepared as SHO.
Export	Scrying Orb	Creates a labelled animated GIF from the processing snapshots.

Core principle

Automation is a recommendation, not final quality control. Visually verify classification, rejected frames, registration, stack borders, gradients and stars. With mosaics and extreme stretches, the true depth pattern of the capture may become visible; not every pattern is a software defect.

Installation, updating and first launch

Download both applications only from the official AstroWizard site and its StackingWizard link. They are portable: run the EXE on Windows, open the extracted app on macOS, or launch the program from the extracted Linux archive. A new build replaces the program file; raw data and saved projects remain separate.

Windows and macOS builds are signed. A very new Windows signature may still trigger SmartScreen. Confirm the official source, choose More info and then Run anyway. Do not use repackaged third-party copies.

AstroWizard first-time setup

- Select GraXpert in Settings & Help and run GraXpert once by itself so its models download.
- Optionally configure StarNet2 for free star removal; on Linux make the executable runnable and keep its weights beside it.
- Select the RC-Astro CLI for BlurXTerminator, NoiseXTerminator and StarXTerminator and verify licensing.
- Choose a UI Scale that keeps all controls visible; Linux may apply the change on restart.
- Set startup update check, Show Tips and Fellowship to preference; settings are remembered.

After a major update, first process a copy of a familiar project. Changes in classification, colour interpretation and stretch are easier to recognise on known data.

File types, linear data and project organisation

StackingWizard chiefly expects FITS subframes. AstroWizard accepts finished FITS, TIFF and XISF stacks. FITS/XISF preserve linear values and headers; 16-bit TIFF is also useful if it was not stretched or clipped beforehand. JPEG is unsuitable as the processing archive because it is normally stretched, compressed and reduced to 8 bit.

Folder/content	Recommended practice	Reason
Lights	Separate target/night; leave originals untouched.	Errors remain reproducible and targets cannot mix.
Darks	Record exposure, gain and temperature.	Thermal signal and amp glow must match.
Flats	One set per filter and unchanged optical train.	Dust and vignette change with rotation, focus and disassembly.
Bias/dark flats	Match gain; dark flats match flat exposure.	Avoids incorrect pedestal/dark correction.
Results	Separate linear, processed, Starless/Stars and receipts.	Return to earlier stages without quality loss.

A finished stack must not be treated as a light among its own subs. Beta 4.4 catches many such cases, but manifest review remains mandatory. Use one project folder per telescope/camera/filter configuration for long projects.

Part I - StackingWizard Beta 4.4

The interface still has four consecutive stages: The Pile, The Looking Glass, The Stack and The Handoff. Beta 4.x chiefly rebuilds the path for walks and mosaics.

Stage	Purpose	Essential check
1. The Pile	Read and sort by type, target, camera, filter and exposure.	Are lights/calibration frames correct? Is Mosaic Auto/One canvas/Panels/Off appropriate?
2. Looking Glass	Grade, play and inspect suspect frames.	Personally check cloud, branches, dew, focus, trails and satellites.
3. The Stack	Calibrate, register, build Live Canvas and perform final vote.	Watch stars, panel placement, depth, warnings, RAM and disk.
4. Handoff	Save, combine sessions and open in AstroWizard.	Choose the correct stack/filter family and archive the linear master.

A. Prepare the data

Keep raw files untouched and use a copy for beta testing. FITS headers take precedence over filenames and folders. Match darks by exposure, gain and preferably temperature; flats require the unchanged optical train, focus and filter. Dark flats belong in Bias / dark flats.

- Convert camera RAW such as NEF/CR2/ARW to FITS first.
- This deep-sky path is not for planets, Moon, Sun or comets.
- Local storage is faster; large sensors need generous RAM and working space.
- Siril-converted subs with stack count one are no longer mistaken for finished stacks.
- A finished stack accidentally mixed with subs, or an extreme exposure outlier, is set aside and named.

Calibration frames remove repeatable sensor and optical defects before registration. Darks model thermal structure and hot pixels. Flats correct pixel response, dust shadows and vignetting. Bias or dark flats supply electronic offset where the calibration path requires it.

Decision rules

- Dark: same exposure and gain/ISO, with temperature as close as possible; take a warning beyond roughly 5 °C seriously.
- Flat: same camera orientation, filter, focus region and optical train; shoot new flats after disassembly.
- Dark flat: same exposure and gain as the flat; do not force an unrelated bias path as well.
- A documented master library is valuable for cooled cameras, but StackingWizard does not yet manage a fully automatic reusable calibration library.

Inspect the linear result for residual amp glow, hot-pixel chains, dust donuts and over/under-corrected vignette. A later gradient process may hide poor calibration visually but cannot repair it physically.

B. The Pile - verify the manifest

Load a project folder or loose files. Verify target, camera, filter, exposure groups and calibration assignments. A flat incorrectly stamped LIGHT can be accepted deliberately after inspection.

Mosaic / walk mode

Mode	Use it when
Auto	Default: discovers named panels, pointing spread and typical walks.
One canvas	Overlapping pointings grow on one Live Canvas, typical for Seestar framing walks.
Panels	Classical panel mosaics; also for Dwarf when each panel has a folder or all headers share coordinates.
Off	A normal strongly dithered single field; no mosaic assembly.

Dwarf note: Dwarf 3/Mini may write one RA/Dec for every panel. Beta 4.2 recognises numbered object names such as "M 31(1)". If Auto cannot infer the set, select Panels.

C. The Looking Glass - grade and inspect

Automatic grading uses star count, bright-star light, softness, trails, noise and sky structure. For a walk, position-dependent criteria compare each frame with its own pointing. Playback resumes where it paused, and grades can be exported as CSV.

- Standard is the sensible starting mood; Lenient retains more borderline frames, Ruthless is stricter.
- Calibration is reserved for stacking; grading sees the exposure as captured.
- Cloud/haze detection does not replace visual review; branches, aircraft, satellites, dew and slow focus drift may survive.
- Round clean stars and a stable background matter more than a perfect score.

Looking Glass grades are decision support. A statistically acceptable frame can still contain a branch, aircraft, drifting dew or a brief tracking jump. Play the sequence at least once and inspect every obvious change in star shape, background or brightness.

Feature	What you see	Decision
Cloud/haze	Fewer stars, weaker bright stars, brighter or patchy sky.	Reject clearly affected frames; weigh marginal ones against lost integration.
Focus/dew	Stars grow gradually, halos increase, contrast falls.	Reject from the visible turning point.
Tracking	Oval/double stars or a directional jump.	Remove isolated faults; fix systematic capture issues.
Satellite/aircraft	Straight trail, blinking sequence or broad track.	Final vote may help, but reject strong frames manually.
Branch	Irregular dark structure, often near an edge.	Remove manually; automation is not reliable here.

Over-aggressive rejection shortens integration and can damage SNR more than modest FWHM variation. Keep a technically usable frame when it adds real depth without an uncorrectable artefact.

D. The Stack - ordinary single field

After review, start the stack. The program calibrates, finds stars, registers and combines the selected frames. Worker count follows available memory. If allocation fails, the final vote divides a band into smaller pieces: bit-identical but slower.

Demosaicing

Bilinear is again the robust default from Beta 4.2. VNG and Malvar-He-Cutler remain deliberate alternatives. VNG may improve star cores but created a fine diagonal maze on one undithered set; always inspect a hard stretch.

Registration finds stars, determines geometry relative to the anchor and resamples every light onto a common grid. In the live view, matched stars should not show systematic edge displacement. Beta 4.4 also demands that a star peak lights neighbouring pixels, preventing warm pixels from becoming anchors.

The final vote combines many samples per pixel and rejects outliers. Too little rejection leaves hot pixels and trails; too much can damage star cores and thinly covered edges. Move the Workbench rejection control only after identifying a real problem.

Inspect the finished stack

- At 100%, check double stars and local distortion.
- Use a hard screen stretch to reveal hot pixels, walking noise, banding and calibration residue.
- Identify black registration borders for clean cropping in AstroWizard.
- For mixed exposures, inspect bright stars for dark boxes or incorrect halo levels.
- [Archive receipts and grade CSV with the stack.](#)

A good linear stack may look unimpressive. Clean stars, even calibration and preserved faint signal matter more than an attractive screen stretch.

E. Live Canvas, walks and mosaics

From Beta 4 every frame lands directly on a growing canvas. Stars determine placement and gold outlines show where each frame landed. All overlaps solve sky levels together while preserving the anchor brightness. Shared stars can also measure and undo field distortion across panel seams.

- Walks are no longer stacked by pointing and stitched afterwards.
- Thin overlaps are judged using stars that could actually land on the existing canvas at the candidate shift.
- A panel meeting the canvas along one narrow seam can keep its level; soft-cut edges avoid dark seams.
- The deepest stack (frames × seconds) becomes the preferred grid when finished stacks are combined.
- One finished frame may serve as a panel; a minimum of three subs is no longer required.

A walk moves through small overlapping pointings during the night. A panel mosaic uses intentionally named or separated fields. Beta 4 builds both on one Live Canvas, but their geometry is not identical.

1. In The Pile verify whether Auto identifies a walk or panels.
2. For Dwarf data with common coordinates choose Panels; use numbered object names or one folder per panel.
3. Watch every outline land plausibly and compare central/edge depth.
4. If a panel is absent, inspect receipt overlap, star count, placement route and held panels.
5. Judge level solving and final vote only after placement is complete.
6. Save the combined mosaic and, for important work, the panel stacks.

Thin overlaps are geometrically fragile. Beta 4.3 tests candidate shifts using stars that could actually land on the placed canvas. A three-panel row no longer needs a two-dimensional spread before the map can predict position.

Unequal depth is not a gradient. If the middle has nine-panel coverage and a corner has one, an extreme stretch must show different noise and depth. Cropping or more exposure is often cleaner than aggressive flattening.

F. Cloud verdict and grid artefacts

Beta 4.4 judges cloud/haze at roughly 100-pixel scale and only on plain sky, excluding bright-star zones and structured nebulosity. A rejected patch must be large and is feathered softly. Thin outer walk pointings drop the single highest and lowest sample at each pixel so a hot pixel or trail does not own the edge.

Under extreme stretch, cells or uneven depth may remain visible. Distinguish true overlap depth from a regular interpolation lattice. The developer explicitly does not claim the issue is universally gone.

G. The Workbench - seven controls for exceptions

The gear beside UI Scale opens the Workbench on every page. The wizard default comes first, with plain-language help. Overrides can last for the session or permanently and are named in receipts.

Control	Change when ...	Usually leave when ...
Cloud verdict	thin cloud remains or too much valid sky is rejected.	the night is clean and background stable.
Final vote rejection	isolated outliers, hot pixels or trails survive.	stars and faint detail are clean.
Per-frame sky shape	walk pointings carry changing vignette/gradient.	good flats exist and it is a single field.
Canvas large gradient	the complete canvas has a slow brightness slope.	real large nebulosity fills the field.
Vignetting from walk	a walk without suitable flats shows optical falloff.	using proper flats or classic panels.
Twin sample on panels	rotated panels show a fine regular lattice.	no rotation-related pattern is visible.
Final-vote memory	8-GB systems or large cameras start swapping.	there is ample RAM.

The Workbench is for measurable exceptions. Document every override, change only one control, build comparisons and judge the same 100% region. Return to Wizard's choice when there is no clear improvement.

Control	How to test	Risk when excessive
Cloud verdict	Compare sky patches and bright-star zones.	Real nebulosity or rare edge coverage is weakened.
Final vote	Inspect hot pixels and trails at 100%.	Star cores and rare fine detail are rejected.
Per-frame sky shape	Hard-stretch individual walk pointings.	Large genuine emission is treated as sky.
Canvas gradient	Compare complete mosaic with depth map.	Astrophysical large-scale nebulosity is flattened.
Walk vignetting	Only without reliable flats and with obvious falloff.	Changing sky becomes a false optical model.
Twin sample	Inspect rotated panels for periodic lattice.	Slightly broader stars and more processing.
Memory	Watch RAM and system swapping.	Too low is slower; too high can swap or fail.

Use session settings for experiments; select forever only after several projects benefit. Reset restores all wizard defaults. Receipts name every non-default choice and are essential for reproducible bug reports.

H. Handoff and archive

- Save the linear FITS stack separately before processing.
- Mono filter stacks share alignment; RGB and narrowband families are not silently mixed.
- The Handoff list scrolls beyond eight stacks so Combine remains reachable.
- Read the receipts: canvas geometry, panel placement, overlap, depth range and every non-default Workbench choice.

Part II - AstroWizard “True Colours”

The core remains a nine-step path. Work from linear to non-linear and keep intermediate files. For an ordinary OSC or smart-telescope colour master, load one finished colour stack. Mono and two-dual-band projects open the appropriate combine route.

Step	Purpose	Recommendation
1 Load	Load linear stack; identify data and combine sets.	Prefer FITS/TIFF/XISF; avoid processed JPEG.
2 Crop	Remove stack borders; rotate and compose.	Remove black registration edges early.
3 Background	Correct gradient/vignetting.	Flatten gently; use GraXpert for complex fields.
4 Colour	Calibrate broadband or design narrowband.	Use Star-True Colours only on suitable linear data.
5 Sharpen	Improve PSF/detail.	Inspect halos and microcontrast.
6 Noise	Reduce random noise.	Keep texture; avoid plastic surfaces.
7 Stretch	Reveal linear signal.	Wizard first; GHS for manual control.
8 Polish	Curves, colour, stars and narrowband.	Several small moves beat one extreme move.
9 Export	Archive, presentation files and GIF.	Keep 16-bit TIFF/FITS; JPEG for delivery.

Step 1 - Load Image

This step defines the data foundation for the entire workflow. Prefer a linear, unstretched FITS, XISF or 16-bit TIFF master. A linear stack looks dark because faint signal still occupies a narrow range; step 7 will reveal it.

Immediate checks

- Is the complete field present and correctly oriented?
- Are black or uneven registration borders visible?
- Is the file truly linear, or did the stacking program already stretch it?
- Is there a gross cast, unusual channel level or clipped highlight?
- Are coordinates, filter and FITS headers plausible? Star-True Colours depends on them.

AstroWizard recognises many Seestar, Dwarf, Siril and PixInsight stack variants. A single two-dimensional Bayer raw frame is identified and refused: calibrate and integrate it in StackingWizard or another stacker first.

Mono and dual-band sets

With several mono masters, filter labels pre-fill Combine Mono Channels. Verify every channel role, especially SHO, HOO and LRGB. Duplicate filters are named and RGB/narrowband families are not silently mixed.

New in True Colours: load one Ha/OIII OSC stack and one SII/OIII stack. The second is registered to the first, Ha/OIII/SII are separated, the two OIII estimates are averaged and SHO is proposed. Verify registration, filter roles and both OIII contributions before committing.

Checkpoint: never overwrite the untouched linear master. Save combinations under new names and retain every input master.

Step 1B - Input data, headers and combinations in detail

Confirm linear state

A linear image is dark with a narrow histogram near the left. Bright stars may show while nebula seems absent. A stretched image already has broad midtones. Treating stretched data as linear breaks colour calibration, sharpening and a second stretch.

Input	Suitability	Check
Linear FITS/XISF	Best: precision and headers retained.	Coordinates, filter, Bayer/channel information and range.
Linear 16-bit TIFF	Very good if exported unchanged.	No display stretch baked in.
Stretched TIFF	Suitable for Final Polish, not full linear route.	Skip linear tools or confirm status.
JPEG/PNG	Only finished or near-finished presentation.	Compression, 8 bit and clipping limit work.
Single Bayer FITS	Not direct input.	Calibrate, register and stack first.

Header and set problems

- Missing RA/Dec blocks Star-True Colours but not other steps.
- Wrong filter labels can mis-map mono; confirm before Combine.
- Unusual TIFF/FITS ranges are normalised, but inspect preview/histogram.
- Large PixInsight headers are tidied for RC-Astro tools.
- Network files may be copied locally for RC-Astro and returned.

Channel and dual-band checks

Before combining, compare stellar geometry, scale, rotation, crop borders and background. OIII in two dual-band stacks should show the same real morphology; major disagreement suggests conditions, gradient or filter-role problems.

If linear state or filter identity is uncertain, stop at step 1. A wrong input classification propagates through every later step.

Step 2 - Crop / Framing

Cropping removes black stack borders, resampling artefacts and poorly covered zones before background or colour tools interpret them as image data. It also establishes rotation and composition.

Controls

- Draw the crop directly on the image.
- Drag inside to move it.
- Drag corners to resize.
- Drag just outside to rotate the image behind the fixed frame.
- Use Fine Rotation for small angular corrections.
- Choose 3:2, 4:3, 2:1 or 9:16 when a fixed output is required.

Auto-fill prevents dead corners after rotation. For walks and mosaics, also inspect depth: an edge covered by only one or two frames may be geometrically valid yet far noisier than the centre.

Composition

Remove technical borders fully while leaving the subject room. Inspect all four corners and avoid cutting extended nebulosity accidentally. Keep a generous master crop, then create a tighter output crop for print or social formats.

Crop and rotation also follow the Starless and star working files, keeping them registered for later recombination.

Step 2B - Framing strategies and edge cases

Crop order

- 1. Identify all black integration borders.
- 2. Judge thin mosaic/walk edges by noise and depth.
- 3. Set rotation without wasting field.
- 4. Compose subject, stellar weight and breathing room.
- 5. Choose final output ratio last.

Lock, tilt and final correction

Newer crop tools include 1:1, 3:2, 4:3, 16:9 and 9:16 locks. Tilt/Fine Rotation can be typed and reset. The exterior remains visible under a shade. Export also offers Last-minute crop for a small late border defect.

Situation	Recommendation	Reason
Dithered single field	Remove only black/unstable borders.	Retain maximum field.
Walk/mosaic	Crop thin-depth outer islands more firmly.	Avoid edge noise and gradient errors.
Large nebula	Generous master crop; target crop later.	Preserve extended faint signal.
Galaxy/small target	Use star field balance; do not always centre.	More natural composition.
Phone/social	Derive 9:16 from high-resolution master.	Archive remains flexible.

Diagnosis

- Dead corners: frame exceeded auto-filled rotation area.
- Star layer mismatch: old version or external geometry change.
- Gradient changes after crop: bad borders influenced the model; recompute Background on cropped master.

Crop is destructive to field area, not to the raw master. Save under a new checkpoint name.

Step 3 - Remove Gradient

This step removes slow brightness and colour trends from light pollution, Moon, vignette or uneven transparency. It must remove technical background, never genuine large-scale nebulosity. Work conservatively and compare continuously.

Flatten the Sky

The built-in method fits a gentle surface to the darkest presumed-sky regions and subtracts tilt or bowl. Two guards decline work when too little clear sky exists or the proposed surface is almost as strong as the sky.

GraXpert

GraXpert is stronger for curved gradients, complex vignette and difficult fields. Run it separately once to download models. The current build can fall back to the newest available model, and on Mac retry on CPU if compilation fails.

Evaluate the result

- Use Hold/Compare and watch extended nebular boundaries.
- Seek a more even background without forcing every area to one value.
- Look for new dark hollows, bright rings or colour patches.
- In nebula-filled fields, accept a residual slope rather than remove real signal.
- Save a linear checkpoint after a successful correction.

SH2-129/OU4 contains broad faint emission. An aggressive model can treat the Squid or diffuse Ha as background. When uncertain, correct less.

Step 3B - Evaluate gradient models and distinguish causes

Feature	Technical gradient	Real nebulosity
Shape	Smooth tilt, bowl, edge glow or colour wedge.	Filamentary, cloudy, object-related.
Sensor relation	Often tied to edge, orientation or light source.	Fixed in sky coordinates.
Channels	May differ by channel but stay spatially smooth.	Ha/OIII/SII have real different morphology.
Crop/rotation	Remains image/sensor related.	Object remains sky related.

Flatten versus GraXpert

- Flatten: fast and repeatable for slow tilt/vignette.
- GraXpert: stronger for complex curved structure; inspect model/AI result.
- With little clear sky, a weaker residual is safer than perfect flatness.
- Ask whether the removed model looks like illumination or nebula.

Artefacts

- Dark hollows around nebula: object entered model.
- Bright/dark corners: vignette or flat overcorrection.
- Colour patches: channel residuals or aggressive model.
- Mosaic seam: often stack/depth issue, not ordinary gradient.

Do not repeat Background with escalating strength. If a clean first attempt fails, inspect calibration, mosaic levels or stray light. Repeated subtraction slowly destroys large-scale signal.

For difficult fields, save a version with a small honest residual gradient. It is preferable to lost faint signal.

Step 4 - Balance Colour

Colour has different goals by data type. Broadband RGB can be tied to physical stellar colours. Dual-band and SHO/HOO are emission-line mappings whose palette is a conscious choice. Select the correct route first.

Star-True Colours - new

For a linear FITS with sky coordinates, AstroWizard fetches several hundred ESA Gaia field stars. One click detects and matches scale, rotation and mirror without an external solver, measures R/G/B and fits camera response. A Sun-like star becomes white and the sky neutral.

- Requires linear broadband data, valid RA/Dec and at least 30 clean stars.
- Saturation, strong halos or sparse fields may prevent calibration.
- The tool honestly declines stretched data, narrowband palettes, missing coordinates or too few matches.
- After calibration, retain a range of red, yellow and blue stars; not every star should be white.

Manual and narrowband colour

Dual-band uses HOO/Narrowband controls for Ha/OIII balance; SHO maps SII, Ha and OIII deliberately. Do not neutralise real emission colour. For L-Ultimate SH2-129/OU4 data, Gaia cannot define an objective nebular palette; stars may be handled separately.

A neutral background is not a colourless nebula. Sample outside the emission field and avoid global moves that make faint OIII green, grey or invisible.

Step 4B - Colour decisions, verification and exceptions

Verify Star-True Colours

- Use only linear broadband.
- Thirty clean matches is a minimum; broad distribution is better.
- A solar white point does not make every star white.
- Judge background, bright stars and faint object colour separately.
- Correct strong gradients first.

Cause of decline	Action
No coordinates	Add valid solved headers externally or use manual route.
Too few stars	Use wider field, fewer saturated stars or manual balance.
Stretched data	Return to linear master.
Narrowband detected	Build HOO/SHO palette deliberately.
Severe saturation	Use better capture/stack or manual colour.

Manual broadband

When Gaia is unavailable, sample genuinely empty background, limit global channel moves and preserve varied star colours. Auto Colour Balance is bounded near $\pm 12\%$ so genuinely warm fields are not pushed into green.

Dual-band and mono

For HOO/SHO, separate tasks: calm background, deliberate emission balance and natural or separately processed stars. Verify LRGB filter roles and SHO S/H/O mapping rather than forcing an objective white point.

Colour is complete when background, stars and subject are each plausible. One global metric cannot always satisfy all three.

Step 5 - Sharpen

Sharpening improves stellar profiles and non-stellar detail but cannot invent signal. Apply it to linear, background-corrected data and judge both at 100% and in the full composition.

Native High Pass

The built-in path is fast and independent. Use small strengths; fine filaments should become clearer without amplifying noise, rims or harsh contours.

BlurXTerminator

BlurX estimates and corrects the PSF and can treat stars and non-stellar detail separately. Custom controls include stellar sharpening, star halos, non-stellar detail and manual PSF diameter. Use manual PSF only when the automatic estimate is demonstrably wrong.

- Inspect bright and medium stars for dark rings.
- Avoid needle-like cores.
- Do not make filaments brittle or surrounded by micro-halos.
- Prefer moderation when star quality varies across the field.
- Sharpen once deliberately: do not turn later contrast tools into a second deconvolution.

If stars look artificial, first reduce stellar strength. Do not conceal excessive sharpening with more halo processing.

Step 5B - Tune BlurX/High Pass and avoid artefacts

Prerequisites

- Linear image, corrected background, not already sharpened repeatedly.
- Enough stars and no major registration error.
- 100% view with bright, medium and small stellar samples.
- Pre-sharpen checkpoint saved.

Control	Increase when ...	Reduce when ...
Stellar sharpening	Stars are soft but clean and round.	Cores become hard or ringed.
Star halos	Halos dominate and are plausibly correctable.	Transitions look cut or unnatural.
Nonstellar detail	Real nebula/galaxy detail is soft.	Noise/pixels become brittle structure.
Manual PSF	Automatic diameter is clearly wrong.	No measured reason exists.

High Pass

High Pass is less model-based and useful as light microcontrast. Keep strength low. Do not automatically combine High Pass and BlurX; choose a primary route and justify any second move.

Failure modes

- Dark rings: too much stellar/halo correction.
- Wormy nebula: noise sharpened as detail.
- Uneven corners: optical/registration issue, not a global strength target.
- Smaller but colourless stars: cores/channels compressed too far.

Sharpening succeeds when detail is clearer but processing is not the first thing the viewer notices.

Step 6 - Reduce Noise

Denoising works best before a strong stretch separates the noise visually. The goal is a calmer surface with fine structure intact, not a perfectly smooth plastic field. Inspect faint nebulosity and background separately.

GraXpert Denoise

GraXpert is an accessible AI route. Run the program separately once so its model is available. Small stars and linear filaments must not disappear or merge into broad patches.

NoiseXTerminator

NoiseX provides finer control. The corrected Custom path sends equal fine/coarse intensity-colour values as plain denoise, and distinct values as band controls, avoiding conflicting CLI switches.

- Begin at moderate overall strength.
- Judge colour and luminance noise separately.
- Inspect at 100%, then return to full-image scale.
- Do not denoise faint OIII until it has no structure.
- Save the final linear pre-stretch master.

Waxy or blotchy nebular edges indicate too much denoise. Undo and restart more gently instead of adding artificial grain later.

Step 6B - Fine-tune NoiseX/GraXpert and verify

Noise appearance	Likely cause	Goal
Fine luminance grain	Photon/read noise.	Moderate smoothing with fine detail.
Colour speckles	Chroma or channel imbalance.	Target colour component.
Large blotches	Gradient, walking noise, low dither or earlier overprocessing.	Investigate stack/background, not just denoise.
Banding/walking	Systematic pattern.	More dither/integration; denoise is insufficient.

NoiseX Custom

NoiseX can separate intensity/colour and fine grain/coarse blotch. Scale sets the transition and Passes repeats work. Do not raise everything together: begin with overall strength, then adjust only the remaining component.

GraXpert

GraXpert is a simple total route. Model and Compute Device can be set in Settings. CPU is slower but valuable after GPU problems. A long CPU run is not necessarily a hang; inspect console/progress.

Quality control

- At 100%, compare stars, thin filaments and faint edges.
- At 50%/Fit, seek calm without waxiness.
- Compare Original: real detail stays fixed while random grain falls.
- Inspect channels for monochrome islands or washed stellar colour.
- Use one main linear denoise; later corrections sparingly.

If coarse blotches become clearer after denoise, return to Background or Stack instead of increasing strength.

Step 7A - Stretch Wizard: selection and preview

Stretch Wizard separates two decisions: colour handling, then stretch depth. Every preview is rendered from the real image with synchronised zoom and pan, so the same dust lane, star and nebular edge can be compared.

See both, pick one: Linked or Unlinked

Mode	Effect	Typical use	Risk
Linked	One transfer relationship for all channels; original ratios remain.	Recommended starting point for broadband, dual-band and narrowband.	A technical cast remains visible and needs proper correction.
Unlinked	Each channel follows its own distribution; background often looks neutral quickly.	Problem channel spread or a deliberate palette after visual comparison.	Real ratios, stellar colour and faint OIII can change.

Current advice is to test Linked first, even for dual-band. The old rule “dual-band always unlinked” is obsolete; field tests often found deeper faint extensions and subtler colour with Linked.

Wizard presets

Preset	Character	Use
Soft / low Custom	Gentle opening with large reserve.	Bright or star-rich fields and cautious first passes.
Normal	Strong universal current default.	Well-calibrated ordinary deep-sky data.
Aggressive	Stronger midtone lift.	Deep data when Normal stays flat.
Deep	Data-driven black/white point and strength.	Very faint dust, halos and nebulae.
Deep - Keep Colours	Deep reach while preserving field colour balance.	Broadband fields with already convincing colour.
Custom	Logarithmic strength with Normal/Aggressive marks.	Intermediate repeatable tuning.

Deep reveals residual gradients, flat errors and blotchy calibration as honestly as faint signal. Fix linear causes rather than crushing the background afterwards.

Keep this look commits the exact preview. First compare background, faint subject and stellar cores at synchronised zoom.

Step 7B - Legacy stretches and manual GHS

Legacy options retain classic and Deep methods. Advanced adds complete manual Generalized Hyperbolic Stretch: not merely a stronger slider, but controlled transformation of selected brightness regions.

GHS controls

Control	Meaning	Practical effect
Transformation	Generalized Hyperbolic or Linear.	GHS shapes selected ranges; Linear changes values more evenly.
Stretch Factor	Overall transformation strength.	Higher opens more; reduce sharply on later passes.
Local Intensity	Local shape/steepness, including negative.	Controls concentration around Symmetry Point.
Symmetry Point	Brightness around which the transformation is centred.	Target background, nebula or brighter structure; type, slide or click image.
Shadow Protection	Protects dark values.	Controls grey background but can remove faint signal.
Highlight Protection	Protects bright values.	Preserves stellar cores and bright nebula.
Invert	Reverses transformation direction.	Special-purpose only.

Read histogram and transfer curve

The left foot holds background/faint signal, the broad middle contains nebula/galaxies, and the right tail holds bright stars and cores. Zoom until background peak and symmetry point are distinct. The curve shows which values expand or compress.

- Symmetry just right of background opens faint nebula while retaining control.
- Symmetry on subject separates midtone filaments.
- Farther right targets brighter structure; watch highlight protection.
- Too much shadow protection looks clean but removes faint outskirts.
- Too little highlight protection turns stars white.

GHS is cumulative. Record Factor, Local Intensity, Symmetry Point and protection values for important projects.

Step 7C - Multi-pass stretch: recipes and failure modes

Safe three-pass workflow

- 1. Open: Linked, moderately strong pass just above background with highlight protection.
- 2. Structure: lower Factor and move Symmetry to relevant nebulosity; inspect stars at 100%.
- 3. Refine: very small pass for faint outskirts or brighter cores; do not clamp shadows.
- 4. Undo a bad Apply; Reset Step restarts all of step 7.
- 5. Save a non-linear master before step 8.

Target	Strategy	Control
Bright emission nebula	Normal/moderate Custom; few GHS passes.	Do not burn stellar cores or bright Ha.
Faint OIII	Linked, useful first opening, then small OIII-range passes.	Noise often grows faster than OIII.
Galaxy/bright core	Moderate stretch, strong highlight protection, separate midtones.	Keep core and tidal arms.
Dust/IFN	Deep or GHS just above background.	Flat, gradient and denoise blotches.
Star cluster	Soft/Normal, preserve colour, few passes.	Avoid swollen white stars.

Common failures

- Grey flat background: overly deep stretch or unresolved linear gradient.
- Black background: shadow protection/black point clipped faint signal.
- White stars: highlight protection too late; go back instead of inventing colour.
- Mint/grey OIII: Unlinked or colour neutralisation shifted channel relationships.
- Noise dominates: stop stretching and inspect integration. gradient and denoise.

For SH2-129/OU4, more global stretch does not guarantee more Squid. Selective OIII work in step 8 is often better.

Step 8A - Curves: tone, colour and masking

Curves is the central finishing tool. Dragging renders the dialog preview; pausing computes sharp full resolution in the background, with multi-core commit.

Channel	Effect	Use
RGB / luminance	Brightness and contrast.	Gentle S-curve, lift midtones, slightly lower background.
R / G / B	Brightness-dependent channel changes.	Correct casts or palette in shadows/midtones/highlights.
S - Saturation	True saturation curve with stable luminance.	Lift weak colour while protecting strong colour.
Colour family	Limits colour curve to selected family.	Target OU4 blue/cyan without moving red Ha.

Controlled S-curve

- Anchor near background.
- Lift midtones slightly.
- Stabilise upper range to protect stellar cores.
- Commit small curves and reassess rather than building an extreme shape.
- Use Hold: Compare Original in preview and inspect the main image at 100% and Fit.

Masked Saturation is more selective than global saturation. Custom Strength 1.0× is the classic effect; start below and inspect background noise and halos.

Saturation cannot create missing OIII. If only chroma noise strengthens, reduce it and use a narrower colour family or mask.

Step 8B - Narrowband Tune, palettes and Colour Spells

Narrowband Tune controls Ha/OIII/SII balance only when those lines exist. AstroWizard warns on a broadband profile because a dual-band palette would drain real colour.

Narrowband method

- Create a neutral starting balance without losing faint OIII.
- Reduce Ha dominance before global saturation.
- Lift nebular OIII/cyan while limiting background cyan.
- For SHO, verify mapping and develop gold/blue gradually.
- Zoom preview across stars, background and weakest nebula.
- Use several small commits.

Spell	Effect	Note
Yellow to Copper & Gold	Faint yellow becomes copper; bright remains gold.	Classic SHO direction; inspect stars/background.
Blue to Teal	Blue shifts toward OIII teal.	Do not turn the whole sky cyan.
Dynamic Pop	Gentle finishing S-curve.	Very cautious after strong contrast.
Add Warmth	Warmer global character.	Stackable; watch cumulative effect.
Cool Down	Cooler character.	Watch stellar colour and Ha/gold.

Spells are stackable, queued off the UI thread, and open through the book icon as editable Curves recipes. Use them as [learning templates](#), not mandatory presets.

For SHO: Narrowband Tune first, at most one suitable spell, then small manual Curves. Strong stacked spells quickly become neon.

Step 8C - Separate, process and replace stars

Star separation lets nebula and stars receive different treatment. Use StarXTerminator or free StarNet2; StarNet2 needs its weights and executable permissions on Linux.

Remove Stars

- Save a non-linear master first.
- Inspect residuals, holes, rings and damaged nebulosity at 100%.
- Process starless only enough for a natural recombination.
- Do not alter geometry; current builds synchronise crop, flip, rotate and bin.
- Save Starless and Stars Only at high bit depth.

Replace Stars

Restore 1.0 gives exact full replacement. Lower values reduce star weight; extremes look punched-out or pasted-on. Inspect small, medium and bright stars separately.

Problem	Cause	Correction
Rings/holes	Incomplete separation or strong starless curve.	Review separation, reduce processing, softer mix.
Pasted stars	Nebula/background changed too far or stars too hard.	Reduce starless contrast, raise restore, match transitions.
Lost star colour	Earlier stretch/saturation or bright star layer.	Control star colour separately and protect highlights.
Artefacts at star sites	Residuals sharpened/saturated.	Clean at 100% or process less aggressively.

Star separation is optional. A gentle integrated curve may be more professional when the original star field is already harmonious.

Step 8D - Recommended sequences and quality control

Broadband finish

- 1. Small luminance S-curve.
- 2. Saturation curve lifts weak colour and protects strong colour.
- 3. Inspect stellar colour and galaxy core.
- 4. Optional Dynamic Pop at reduced strength.
- 5. Final Compare Original and master save.

Dual-band HOO / SH2-129-OU4

- 1. Set Ha/OIII balance in Narrowband Tune.
- 2. Selectively lift OU4 blue/cyan while protecting background/stars.
- 3. Shape red Ha gently, not with global oversaturation.
- 4. Optionally separate stars, process nebula modestly and replace naturally.
- 5. At Fit: Squid visible but not neon; background neutral; Ha structured.

SHO finish

- 1. Verify mapping and initial neutrality.
- 2. Tune basic gold/blue ratio.
- 3. Use Copper & Gold or Blue to Teal once as a starting point.
- 4. Open recipe and adapt in Curves.
- 5. Apply Masked Saturation/Make it Pop modestly.

Stop criteria

- Background keeps natural fine texture without chroma patches.
- Faint nebula is visible without crushed blacks.
- Stars retain varied colour, soft halos and no dark rings.
- Contrast leads the eye to the subject, not processing artefacts.
- Another move should improve at least two regions without clearly harming a third.

Step 8 is complete when the picture is balanced, not when every tool has been used. Save variants rather than overwriting earlier good states.

Step 9 - Export & Share

Export is part of quality control. Save more than a small JPEG so the project remains open to future tools, printing and variants.

File	Recommendation	Purpose
Linear master	FITS/XISF, full precision, pre-stretch.	Complete reprocessing and future software.
Pre-stretch master	FITS/XISF after background, sharpen and denoise.	New stretch without repeating linear work.
Final master	16-bit TIFF or FITS/XISF.	Archive, print and external finishing.
Starless / Stars Only	Separate high-bit files.	Future star variants and recombination.
JPEG	sRGB, target dimensions, moderate compression.	Web and messaging.
Receipts / notes	Keep with project and version.	Reproducibility and diagnosis.

Scrying Orb - new

Scrying Orb makes an animated GIF from saved snapshots: one labelled frame per step, longer first/last holds and 720-1440 pixel width. Branding is optional. Check size, legibility and distracting brightness jumps.

Final verification

- Reopen the exported file rather than trusting preview alone.
- Verify dimensions, bit depth and colour space.
- Check histogram for clipped shadows/highlights.
- Inspect JPEG for banding, blocks and excessive saturation.
- Name by object, processing state, date and palette.

The high-quality master is the archive; JPEG is only a delivery copy. Do not discard linear and 16-bit checkpoints after a web export succeeds.

Troubleshooting and final checklist

Symptom	Likely cause	Action
Regular mosaic grid	Interpolation, cloud cells or real depth changes.	Restack in 4.4, test Twin sample, compare hard stretch with depth; send receipts if repeatable.
Dark box at bright star	Mixed exposures / old cloud logic.	Restack in 4.4; inspect exposure headers and Cloud verdict.
Panel missing	Insufficient overlap or wrong mode/name.	Force Panels; inspect names and receipt geometry.
Stack appears colour-cast	Preview stretch or wrong Bayer pattern.	Check Colour door and linear values, not preview alone.
Nebula disappears after Background	Model captured real signal.	Undo; weaken Flatten or use controlled GraXpert.
Star-True Colours unavailable	Narrowband, stretched, no coordinates or <30 stars.	Use the appropriate manual colour route; do not force it.
Dark star rings	Excess stellar sharpening/halo control.	Reduce BlurX strength; moderate halo handling.
GHS ruins background	Step too large or wrong symmetry point.	Undo; smaller Apply passes and shadow protection.

Final checklist

- Manifest and calibration assignment are plausible.
- Suspect frames were visually inspected, not only scored.
- No double stars, major calibration residue, unexplained seams or missing panels.
- Linear FITS master archived separately.
- Background correction preserves extended nebulosity.
- Colour method matches the data: Gaia/broadband or deliberate narrowband palette.
- Sharpening has no halos; denoising has no plastic texture.
- Stretch preserves faint signal and star colour.
- 16-bit master and delivery copy saved; receipts/CSV archived for important projects.

Official sources

AstroWizard: <https://astrowizard.lukomatico.com/>

Changelog: <https://astrowizard.lukomatico.com/changelog.html>

StackingWizard: follow “Get the beta” from the AstroWizard site.

Source status: 18 September 2026. Gaia star data: ESA Gaia mission, processed by DPAC. GHS mathematics follows Dave Payne’s published transformation family; AstroWizard’s implementation is independent.